# THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

## ANSWER KEY & MARKING SCHEME SET 1

Chapters 5–8: **Circles, Perimeter and Area, Probability, Sequences and Progressions** | Class 9 Maths | Maximum Marks: 50

### SECTION — A (Objective Type) — Answers — 15 × 1 = 15

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(c)** 48 | Chord $= 2\sqrt{r^2 - d^2}$. |
| 2 | **(b)** 84 | Use Heron's formula. |
| 3 | **(c)** $\frac{11}{50}$ | Relative frequency of the outcome. |
| 4 | **(d)** 53 | $2 \times 25 + 3$. |
| 5 | **(d)** 15 | $d = \sqrt{r^2 - (\text{half-chord})^2}$. |
| 6 | **(d)** 21 | $s = \frac{a+b+c}{2}$. |
| 7 | **(c)** $\frac{7}{25}$ | Relative frequency. |
| 8 | **(b)** 17th term | $3n + 2 = 53$ gives $n = 17$. |
| 9 | **(d)** $37^\circ$ | Half of the central angle. |
| 10 | **(a)** 616 | Find $r$ from $C = 2\pi r$, then use $A = \pi r^2$. |
| 11 | **(d)** $\frac{5}{9}$ | Count vowels among all the letters (repeats counted). |
| 12 | **(a)** 19 | AP with $a = 4$, $d = 3$: $t_6 = 4 + 5 \times 3$. |
| 13 | **(c)** $147^\circ$ | Angles on the two arcs are supplementary: $180^\circ - t$. |
| 14 | **(b)** 176 | Distance in one turn = circumference. |
| 15 | **(c)** $\frac{3}{8}$ | Use the 8 equally likely outcomes. |

### SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 16 | Half-chord $= \sqrt{26^2 - 10^2} = 24$ cm (1); Chord $= 48$ cm (1). | 1+1 |
| 17 | $s = 21$ cm (1); $A = \sqrt{s(s-a)(s-b)(s-c)} = 84$ cm$^2$ (1). | 1+1 |
| 18 | Even outcomes: $2, 4, 6$ with frequencies $22, 16, 16$ (1); $P = \frac{54}{100} = \frac{27}{50}$ (1). | 1+1 |
| 19 | $t_8 = 25$ (1); $3n + 1 = 37$ gives $n = 12$, so yes (1). | 1+1 |
| 20 | Angle on the major arc $= 30^\circ$ (1); Angle on the minor arc $= 180^\circ - 30^\circ = 150^\circ$ (1). | 1+1 |

### SECTION — C (Short Answer) — Answers — 5 × 3 = 15

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 21 | $OA = OB$ so $\angle OAB = \angle OBA$ (1); $\angle OAB = \frac{180^\circ - 70^\circ}{2} = 55^\circ$ (1); Angle on the major arc $= \frac{70^\circ}{2} = 35^\circ$ (1). | 1+1+1 |
| 22 | $s = 21$ cm (1); $A = 84$ cm$^2$ (1); $h = \frac{2A}{15} = \frac{168}{15} = \frac{56}{5}$ cm (1). | 1+1+1 |
| 23 | Total $= 12$ (1); $P(\text{red}) = \frac{1}{3}$ (1); $P(\text{not red}) = \frac{2}{3}$; $P(\text{black or white}) = \frac{2}{3}$ (1). | 1+1+1 |
| 24 | $7 + (n-1)(6) = 151$ (1); $(n-1) \times (6) = 144$ (1); $n = 25$ (1). | 1+1+1 |
| | **OR:** $\angle A + \angle C = 180^\circ$; $5x + 5 = 180$ (1); $x = 35$ (1); $\angle A = 80^\circ$ (1). | 1+1+1 |
| 25 | $s = 13$ cm (1); $A = \sqrt{(s-a)(s-b)(s-c)(s-d)}$ (1); $A = 36$ cm$^2$ (1). | 1+1+1 |
| | **OR:** Walk $= 200 - 80 - 50 = 70$ (1); $P(\text{walk}) = \frac{7}{20}$ (1); $P(\text{not bus}) = 1 - \frac{2}{5} = \frac{3}{5}$ (1). | 1+1+1 |

### SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 26 | Distance of first chord from centre $= \sqrt{13^2 - 5^2} = 12$ cm (1); Distance of second chord $= \sqrt{13^2 - 12^2} = 5$ cm (1); Same side: subtract the distances (1); Distance between chords $= \lvert 12 - 5 \rvert = 7$ cm (2). | 1+1+1+2 |
| | **OR:** Outer area $= \frac{22}{7} \times 21^2 = 1386$ m$^2$ (1); Inner area $= 616$ m$^2$ (1); Track area $= 770$ m$^2$ (1); Cost $= 770 \times 5$ (1); $=$ ₹3850 (1). | 1+1+1+1+1 |
| 27 | Total trials $= 100$ (1); Prime: $2, 3, 5$: $\frac{53}{100} = \frac{53}{100}$ (1); Greater than 4: $\frac{30}{100} = \frac{3}{10}$ (1); Less than 3: $\frac{37}{100} = \frac{37}{100}$ (1); All six frequencies add to 100, so relative frequencies add to 1 (1). | 1+1+1+1+1 |

| | | |
|---|---|---|
| | **OR:** Fares ₹240, ₹280, ₹320 (1); Common difference ₹40, so an AP (1); $t_n = a + (n-1)d$ with $a = 240$, $d = 40$ (1); $t_{12} = 240 + 11 \times 40$ (1); Fare ₹680 (1). | 1+1+1+1+1 |

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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## ANSWER KEY & MARKING SCHEME SET 2

Chapters 5–8: **Circles, Perimeter and Area, Probability, Sequences and Progressions** | Class 9 Maths | Maximum Marks: 50

### SECTION — A (Objective Type) — Answers — 15 × 1 = 15

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(c)** 47 | $a + (n-1)d = 5 + 14 \times (3)$. |
| 2 | **(b)** $55^\circ$ | The exterior angle equals the interior opposite angle. |
| 3 | **(d)** 70 | Area $= \frac{1}{2}d_1d_2$. |
| 4 | **(a)** $\frac{3}{4}$ | Count the balls that are not black. |
| 5 | **(d)** 210 | $S_n = \frac{n}{2}[2a + (n-1)d]$ with $a = 3, d = 4$. |
| 6 | **(b)** $50^\circ$ | $OA = OB$ (radii) so the base angles are equal. |
| 7 | **(d)** 36 | Use Brahmagupta's formula $\sqrt{(s-a)(s-b)(s-c)(s-d)}$. |
| 8 | **(b)** 70 | $P(\text{not } E) = 1 - P(E)$. |
| 9 | **(d)** 162 | $ar^{n-1} = 2 \times 3^4$. |
| 10 | **(a)** equidistant | Recall the theorem. |
| 11 | **(a)** 44 | Arc $= \frac{\theta}{360} \times 2\pi r$. |
| 12 | **(a)** $\frac{1}{3}$ | Count favourable outcomes out of 6. |
| 13 | **(a)** 21 | $3 + (n-1)4 = 83$ gives $n = 21$. |
| 14 | **(c)** 14 | Chord $= 2\sqrt{r^2 - d^2}$. |
| 15 | **(c)** 22 | Perimeters are equal. |

### SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 16 | Circumference $= 176$ cm (1); Revolutions $= \frac{4400}{176} = 25$ (1). | 1+1 |
| 17 | Number of letters $= 11$ (1); $P = \frac{7}{11}$ (1). | 1+1 |
| 18 | Differences $= 5$, constant (1); $t_{12} = 8 + 11 \times (5) = 63$ (1). | 1+1 |
| 19 | Exterior angle $=$ interior opposite angle (1); $\angle DCE = 62^\circ$ (1). | 1+1 |
| 20 | Arc $= \frac{120}{360} \times 2\pi r$ (1); $= 44$ cm (1). | 1+1 |

### SECTION — C (Short Answer) — Answers — 5 × 3 = 15

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 21 | $a = 6$, $d = 5$ (1); $S_n = \frac{n}{2}[2a + (n-1)d]$ (1); $S_{15} = 615$ (1). | 1+1+1 |
| 22 | $r = \sqrt{15^2 + 8^2} = 17$ cm; Right triangle legs 15 cm and 8 cm (1); Area $= \frac{1}{2} \times 15 \times 8$ (1); $= 60$ cm$^2$ (1). | 1+1+1 |
| 23 | Parallelogram: $A = 14 \times 9 = 126$ cm$^2$ (1); Triangle: $\frac{1}{2} \times$ base $\times 18 = 126$ (1); base $= 14$ cm (1). | 1+1+1 |
| 24 | $P(2 \text{ heads}) = \frac{24}{100} = \frac{6}{25}$ (1); $P(1 \text{ head}) = \frac{52}{100} = \frac{13}{25}$ (1); $P(0 \text{ heads}) = \frac{24}{100} = \frac{6}{25}$ (1). | 1+1+1 |
| | **OR:** $r = 4$ (1); $t_n = ar^{n-1} = 3 \times 4^4$ (1); $t_5 = 768$ (1). | 1+1+1 |
| 25 | $OA = OB$ so $\angle OAB = \angle OBA$ (1); $\angle OAB = \frac{180^\circ - 90^\circ}{2} = 45^\circ$ (1); Angle on the major arc $= \frac{90^\circ}{2} = 45^\circ$ (1). | 1+1+1 |
| | **OR:** $s = 24$ cm (1); $A = 84$ cm$^2$ (1); $h = \frac{2A}{21} = \frac{168}{21} = 8$ cm (1). | 1+1+1 |

### SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 26 | $\angle C = 180^\circ - 75^\circ = 105^\circ$ (1); $\angle D = 180^\circ - 88^\circ = 92^\circ$ (1); $\angle DCE = \angle A = 75^\circ$ (1); $\angle DAF = 180^\circ - 75^\circ = 105^\circ$ (1); Reasons: opposite angles supplementary; exterior angle property (1). | 1+1+1+1+1 |
| | **OR:** Outer length $= 34$ m, outer breadth $= 24$ m (1); Outer area $= 816$ m$^2$ (1); Lawn area $= 600$ m$^2$ (1); Path area $= 216$ m$^2$ (1); Units stated (1). | 1+1+1+1+1 |
| 27 | Number of letters $= 11$ (1); $P(\text{vowel}) = \frac{4}{11}$ (1); $P(S) = \frac{4}{11}$ (1); $P(A \text{ to } M) = \frac{5}{11}$ (1); Repeated letters are counted each time (1). | 1+1+1+1+1 |

| | | |
|---|---|---|
| | **OR:** Monthly savings form an AP (1); $t_{12} = 300 + 11 \times 50 = 850$ (1); $S_n = \frac{n}{2}[2a + (n-1)d]$ (1); $S_{12} = \frac{12}{2}[600 + 550]$ (1); Total ₹6900 (1). | 1+1+1+1+1 |

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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## ANSWER KEY & MARKING SCHEME SET 3

Chapters 5–8: **Circles, Perimeter and Area, Probability, Sequences and Progressions** | Class 9 Maths | Maximum Marks: 50

### SECTION — A (Objective Type) — Answers — 15 × 1 = 15

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(c)** $0 \le P(E) \le 1$ | Recall the definition. |
| 2 | **(a)** an arithmetic progression | Definition of an AP. |
| 3 | **(a)** 20 | $d = \sqrt{r^2 - (\text{half-chord})^2}$. |
| 4 | **(b)** 42 | Use Heron's formula. |
| 5 | **(c)** $\frac{7}{50}$ | Relative frequency of the outcome. |
| 6 | **(b)** 44 | $3 \times 16 - 4$. |
| 7 | **(b)** $44^\circ$ | Half of the central angle. |
| 8 | **(c)** 18 | $s = \frac{a+b+c}{2}$. |
| 9 | **(b)** $\frac{9}{40}$ | Relative frequency. |
| 10 | **(b)** 21th term | $4n - 3 = 81$ gives $n = 21$. |
| 11 | **(d)** $131^\circ$ | Angles on the two arcs are supplementary: $180^\circ - t$. |
| 12 | **(d)** 1386 | Find $r$ from $C = 2\pi r$, then use $A = \pi r^2$. |
| 13 | **(a)** $\frac{1}{3}$ | Count vowels among all the letters (repeats counted). |
| 14 | **(d)** 29 | AP with $a = 9$, $d = 4$: $t_6 = 9 + 5 \times 4$. |
| 15 | **(c)** $68^\circ$ | The exterior angle equals the interior opposite angle. |

### SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 16 | Sample space has 8 outcomes (1); $P = \frac{3}{8}$ (1). | 1+1 |
| 17 | $t_2 = 3$ (1); Terms $1, 3, 7, 15, 31$ (1). | 1+1 |
| 18 | Half-chord $= \sqrt{26^2 - 24^2} = 10$ cm (1); Chord $= 20$ cm (1). | 1+1 |
| 19 | $s = 21$ cm (1); $A = \sqrt{s(s-a)(s-b)(s-c)} = 42$ cm$^2$ (1). | 1+1 |
| 20 | Even outcomes: $2, 4, 6$ with frequencies $12, 15, 16$ (1); $P = \frac{43}{100} = \frac{43}{100}$ (1). | 1+1 |

### SECTION — C (Short Answer) — Answers — 5 × 3 = 15

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 21 | Total $= 15$ (1); $P(\text{red}) = \frac{2}{5}$ (1); $P(\text{not red}) = \frac{3}{5}$; $P(\text{black or white}) = \frac{3}{5}$ (1). | 1+1+1 |
| 22 | $90 + (n-1)(-5) = 25$ (1); $(n-1) \times (-5) = -65$ (1); $n = 14$ (1). | 1+1+1 |
| 23 | $\angle A + \angle C = 180^\circ$; $5x + 5 = 180$ (1); $x = 35$ (1); $\angle A = 85^\circ$ (1). | 1+1+1 |
| 24 | $s = 24$ cm (1); $A = \sqrt{(s-a)(s-b)(s-c)(s-d)}$ (1); $A = 80$ cm$^2$ (1). | 1+1+1 |
| | **OR:** Walk $= 250 - 100 - 75 = 75$ (1); $P(\text{walk}) = \frac{3}{10}$ (1); $P(\text{not bus}) = 1 - \frac{2}{5} = \frac{3}{5}$ (1). | 1+1+1 |
| 25 | $a = 25$, $d = -3$ (1); $S_n = \frac{n}{2}[2a + (n-1)d]$ (1); $S_{10} = 115$ (1). | 1+1+1 |
| | **OR:** $r = \sqrt{8^2 + 15^2} = 17$ cm; Right triangle legs 8 cm and 15 cm (1); Area $= \frac{1}{2} \times 8 \times 15$ (1); $= 60$ cm$^2$ (1). | 1+1+1 |

### SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 26 | Distance of first chord from centre $= \sqrt{17^2 - 8^2} = 15$ cm (1); Distance of second chord $= \sqrt{17^2 - 15^2} = 8$ cm (1); Same side: subtract the distances (1); Distance between chords $= \lvert 15 - 8 \rvert = 7$ cm (2). | 1+1+1+2 |
| | **OR:** Outer area $= \frac{22}{7} \times 28^2 = 2464$ m$^2$ (1); Inner area $= 1386$ m$^2$ (1); Track area $= 1078$ m$^2$ (1); Cost $= 1078 \times 4$ (1); $=$ ₹4312 (1). | 1+1+1+1+1 |
| 27 | Total trials $= 100$ (1); Prime: $2, 3, 5$: $\frac{51}{100} = \frac{51}{100}$ (1); Greater than 4: $\frac{35}{100} = \frac{7}{20}$ (1); Less than 3: $\frac{30}{100} = \frac{3}{10}$ (1); All six frequencies add to 100, so relative frequencies add to 1 (1). | 1+1+1+1+1 |

| | | |
|---|---|---|
| | **OR:** Fares ₹175, ₹200, ₹225 (1); Common difference ₹25, so an AP (1); $t_n = a + (n-1)d$ with $a = 175$, $d = 25$ (1); $t_{18} = 175 + 17 \times 25$ (1); Fare ₹600 (1). | 1+1+1+1+1 |

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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